

## CANYONLANDS FIELD, UT, USA

WMO: 724776

Lat: 38.750N

Lon: 109.763W

Elev: 4560

StdP: 12.43

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 93075

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |            |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>1.4 | (c)<br>8.6 | (d)<br>-6.4                   | (e)<br>4.6 | (f)<br>18.1 | (g)<br>-2.0 | (h)<br>5.8 | (i)<br>26.8 | (j)<br>22.0           | (k)<br>44.9 | (l)<br>18.5 | (m)<br>37.7 | (n)<br>1.3               | (o)<br>260 | (p)<br>0.542 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>27.8            | (c)<br>99.7     | (d)<br>62.3 | (e)<br>97.7 | (f)<br>60.9 | (g)<br>95.5 | (h)<br>59.8 | (i)<br>66.2         | (j)<br>86.2 | (k)<br>64.9 | (l)<br>85.2 | (m)<br>63.8 | (n)<br>84.6 | (o)<br>9.5           | (p)<br>250 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.1                            | 99.0 | 70.0 | 59.3 | 89.4 | 70.3 | 56.4 | 80.3 | 71.4 | 33.3          | 84.9 | 32.2 | 84.9 | 31.3 | 84.0 | 85.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      |                            | Min  | Max   | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)  | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 25.5              | 21.1 | 18.3 | DB                         | -1.1 | 103.8 | 8.8                | 2.2                                                | -7.4      | 105.4 | -12.6      | 106.6 | -17.5      | 107.8 | -23.9      | 109.4 |
|                   |      |      | WB                         | -1.8 | 70.0  | 8.3                | 4.9                                                | -7.8      | 73.5  | -12.7      | 76.4  | -17.3      | 79.1  | -23.4      | 82.7  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 54.6   | 27.3  | 35.6  | 46.1  | 53.6  | 63.5  | 75.3  | 82.0  | 78.1  | 69.2  | 54.6  | 40.5  | 28.0  |  |  |
|                                                                        | DBStd       | 19.94  | 9.97  | 8.20  | 7.24  | 7.39  | 7.96  | 6.64  | 4.30  | 4.11  | 6.65  | 7.54  | 7.54  | 9.02  |  |  |
|                                                                        | HDD50       | 2337   | 703   | 403   | 162   | 48    | 5     | 0     | 0     | 0     | 0     | 40    | 295   | 681   |  |  |
|                                                                        | HDD65       | 5298   | 1168  | 822   | 585   | 345   | 125   | 9     | 0     | 0     | 35    | 329   | 734   | 1146  |  |  |
|                                                                        | CDD50       | 4015   | 0     | 1     | 42    | 156   | 423   | 759   | 992   | 870   | 578   | 183   | 11    | 0     |  |  |
|                                                                        | CDD65       | 1501   | 0     | 0     | 0     | 4     | 77    | 318   | 527   | 405   | 162   | 8     | 0     | 0     |  |  |
|                                                                        | CDH74       | 21026  | 0     | 0     | 16    | 145   | 1311  | 4814  | 7376  | 5087  | 2111  | 166   | 0     | 0     |  |  |
|                                                                        | CDH80       | 11134  | 0     | 0     | 0     | 19    | 479   | 2665  | 4368  | 2693  | 882   | 28    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.2    | 3.8   | 5.1   | 6.7   | 8.6   | 8.4   | 8.3   | 6.9   | 6.4   | 6.2   | 5.4   | 4.5   | 3.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 7.4    | 0.5   | 0.5   | 0.5   | 0.6   | 0.6   | 0.4   | 0.6   | 0.9   | 0.9   | 0.9   | 0.5   | 0.5   |  |  |
|                                                                        | PrecMax     | 10.2   | 1.7   | 1.2   | 1.3   | 1.7   | 1.6   | 1.6   | 1.5   | 1.7   | 2.9   | 4.7   | 1.3   | 1.6   |  |  |
|                                                                        | PrecMin     | 4.9    | 0.2   | 0.0   | 0.1   | 0.1   | 0.1   | 0.0   | 0.1   | 0.1   | 0.3   | 0.0   | 0.0   | 0.1   |  |  |
|                                                                        | PrecStd     | 1.7    | 0.4   | 0.4   | 0.3   | 0.4   | 0.5   | 0.4   | 0.5   | 0.5   | 0.6   | 1.0   | 0.3   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 54.3  | 63.0  | 76.3  | 82.5  | 93.3  | 101.4 | 103.4 | 99.4  | 95.1  | 83.3  | 68.7  | 54.4  |  |  |
|                                                                        |             | MCWB   | 42.2  | 44.4  | 49.1  | 52.0  | 59.1  | 60.5  | 62.7  | 63.2  | 60.2  | 54.0  | 48.4  | 43.0  |  |  |
|                                                                        | 2%          | DB     | 48.3  | 57.4  | 70.7  | 78.5  | 89.4  | 98.6  | 100.4 | 97.3  | 91.2  | 78.9  | 63.9  | 49.1  |  |  |
|                                                                        |             | MCWB   | 39.5  | 42.4  | 46.7  | 49.9  | 56.0  | 59.7  | 62.8  | 61.1  | 58.6  | 52.6  | 45.6  | 40.1  |  |  |
|                                                                        | 5%          | DB     | 44.9  | 53.5  | 66.3  | 74.6  | 85.2  | 96.2  | 99.0  | 95.0  | 88.6  | 74.5  | 60.6  | 45.3  |  |  |
|                                                                        |             | MCWB   | 37.0  | 40.9  | 44.8  | 48.4  | 53.9  | 58.2  | 62.4  | 60.1  | 57.6  | 51.1  | 44.0  | 36.6  |  |  |
|                                                                        | 10%         | DB     | 40.9  | 49.1  | 62.6  | 71.4  | 81.7  | 93.1  | 96.9  | 92.5  | 85.5  | 70.4  | 55.3  | 41.2  |  |  |
|                                                                        |             | MCWB   | 34.6  | 38.7  | 43.2  | 47.4  | 52.9  | 57.1  | 61.3  | 60.3  | 56.8  | 49.6  | 41.8  | 34.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 44.2  | 46.5  | 50.1  | 53.5  | 61.6  | 64.0  | 72.5  | 67.8  | 63.9  | 58.3  | 50.4  | 45.0  |  |  |
|                                                                        |             | MCDB   | 49.3  | 58.2  | 72.2  | 77.0  | 92.2  | 94.5  | 95.7  | 86.8  | 79.4  | 72.8  | 64.3  | 52.5  |  |  |
|                                                                        | 2%          | WB     | 40.9  | 44.1  | 47.9  | 51.2  | 57.5  | 62.0  | 66.8  | 66.1  | 62.1  | 55.6  | 47.4  | 41.1  |  |  |
|                                                                        |             | MCDB   | 47.2  | 54.4  | 67.1  | 73.0  | 82.4  | 91.3  | 87.6  | 84.4  | 79.0  | 69.5  | 60.2  | 47.6  |  |  |
|                                                                        | 5%          | WB     | 38.0  | 41.9  | 46.0  | 49.8  | 55.5  | 60.5  | 65.3  | 64.9  | 60.7  | 53.6  | 45.0  | 37.5  |  |  |
|                                                                        |             | MCDB   | 43.3  | 51.4  | 63.5  | 70.8  | 79.4  | 88.7  | 86.7  | 83.0  | 78.5  | 67.6  | 57.8  | 43.8  |  |  |
|                                                                        | 10%         | WB     | 35.1  | 39.7  | 44.1  | 48.2  | 53.8  | 59.1  | 64.3  | 63.6  | 59.2  | 51.5  | 42.8  | 34.9  |  |  |
|                                                                        |             | MCDB   | 40.2  | 47.8  | 60.1  | 67.9  | 77.4  | 86.9  | 86.3  | 81.7  | 77.8  | 66.9  | 54.0  | 40.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.8  | 21.2  | 26.0  | 26.8  | 27.7  | 30.2  | 27.8  | 27.3  | 27.8  | 25.9  | 24.1  | 19.8  |  |  |
|                                                                        |             | MCDBR  | 23.3  | 26.9  | 32.9  | 32.4  | 33.3  | 33.5  | 30.6  | 31.4  | 31.6  | 31.4  | 29.6  | 23.4  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.3  | 14.8  | 16.3  | 15.4  | 14.3  | 14.2  | 12.4  | 12.5  | 13.4  | 14.9  | 16.5  | 15.4  |  |  |
|                                                                        |             | MCWBR  | 20.3  | 23.6  | 29.6  | 29.7  | 30.1  | 30.1  | 25.7  | 24.7  | 24.1  | 24.6  | 26.0  | 21.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.254 | 0.265 | 0.284 | 0.308 | 0.317 | 0.316 | 0.364 | 0.368 | 0.324 | 0.292 | 0.264 | 0.253 |  |  |
|                                                                        | taud        |        | 2.561 | 2.534 | 2.483 | 2.417 | 2.442 | 2.445 | 2.357 | 2.355 | 2.465 | 2.523 | 2.556 | 2.571 |  |  |
|                                                                        | Ebn at Noon |        | 301   | 310   | 312   | 307   | 305   | 304   | 288   | 285   | 293   | 294   | 294   | 293   |  |  |
|                                                                        | Edh at Noon |        | 23    | 27    | 31    | 36    | 36    | 36    | 39    | 38    | 32    | 27    | 23    | 21    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 841   | 1126  | 1607  | 2019  | 2320  | 2595  | 2349  | 2069  | 1798  | 1334  | 956   | 724   |  |  |
|                                                                        | RadStd      |        | 61    | 89    | 105   | 99    | 121   | 131   | 108   | 94    | 83    | 90    | 64    | 50    |  |  |

## Historical Trends

|      |                         | Heating    |            | Cooling    |            |              | Degree-Days |            |            |            |            |
|------|-------------------------|------------|------------|------------|------------|--------------|-------------|------------|------------|------------|------------|
|      |                         | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP        | HDD50       | HDD65      | CDD50      | CDD65      |            |
| (46) | Station Only            | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>-1.03 | (i)<br>N/A  | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| (47) | Regional (13 neighbors) | N/A        | N/A        | N/A        | N/A        | N/A          | N/A         | N/A        | N/A        | N/A        | N/A        |

Nomenclature: See separate page